Ares Koumblis

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Robotics — Expected Aug 2028
GPA: 4.33/4.33

National Technical University of Athens — Athens, Greece
Integrated Master's (BE/ME) in Electrical and Computer Engineering — Feb 2024
GPA: 9.3/10.0 (top 3%)

PROFESSIONAL EXPERIENCE

Tesla — Palo Alto, CA
Senior Autopilot Software Development Engineer — Aug 2025 – Jul 2026
Autopilot Software Development Engineer — Feb 2025 – Aug 2025
Autopilot Software QA Engineer — Jan 2024 – Feb 2025

- Fine-tuned end-to-end driving policies with GRPO, reducing speeding in low-visibility conditions without regressing ride comfort
- Enhanced explicit trajectory selection logic, enabling FSD v14 on HW3 vehicles and left-hand-traffic markets
- Implemented reduced-visibility slowdown logic across vehicle platforms, cutting simulated collision rate by 44%
- Led validation of Tesla's closed-loop video-generation simulator, demonstrating strong correlation with both track-test results and real-world miles per critical intervention
- Developed safety and comfort evaluation metrics across Tesla's open-loop, closed-loop, and video-generation simulators, enabling automated rejection of underperforming models and reducing reliance on on-vehicle testing
- Adapted Lane Departure Avoidance for rear-wheel steering and deployed on Cybertruck for 5-star NCAP
- Led platform-specific validation and release efforts for the first FSD deployment on Cybertruck (~30k vehicles)

RESEARCH

Multimodal Embodied Agents: 3D VLMs, Long-Term Memory, and Tool Use — Pittsburgh, PA
Carnegie Mellon University – Katerina Fragkiadaki Lab — Aug 2026 – Present

- Distilled multi-step, tool-calling reasoning traces from a 2D VLM (Qwen3.8-27B) into a 3D VLM (Qwen3D) via SFT, teaching it to actively inspect point clouds (zoom, rotate) for 3D grounding (targeting CVPR 2027)
- Led imitation learning (DAgger) and RL (GSPO) post-training of a 3D VLN policy for long-horizon embodied navigation, improving over our SFT model by 10% on R2R and RxR and 15% on OVON (targeting CVPR 2027)
- Developing learned memory management, where the agent decides what to write, retrieve, and fold out of context, for long-horizon embodied tasks (targeting ICML 2027)

PROJECTS

Autonomous Formula Student (FSAE) Race Car — Athens, Greece
National Technical University of Athens — Oct 2021 – Aug 2023

- Founded and led a 10-member team to develop and deliver an autonomous race car over a 2-year timeline
- Deployed an edge perception stack (YOLOv5 on TPU, ResNet keypoint detector on iGPU) for real-time track cone detection and distance estimation, earning 4th place in the ZEISS Real-Time Video Processing Award
- Developed a YOLOv8 variant with a regression head for direct cone distance estimation via knowledge distillation from the baseline perception stack
- Built a graph-based SLAM system using the iSAM algorithm for real-time cone and vehicle localization
- Trained an encoder-decoder network to model nonlinear vehicle dynamics, deployed in a Koopman MPC for path tracking

SKILLS

Programming Languages: Python, C++, C
ML & Robotics: PyTorch, HuggingFace, Scikit-Learn, OpenCV, GTSAM
Developer Tools: Git, Linux, Weights & Biases